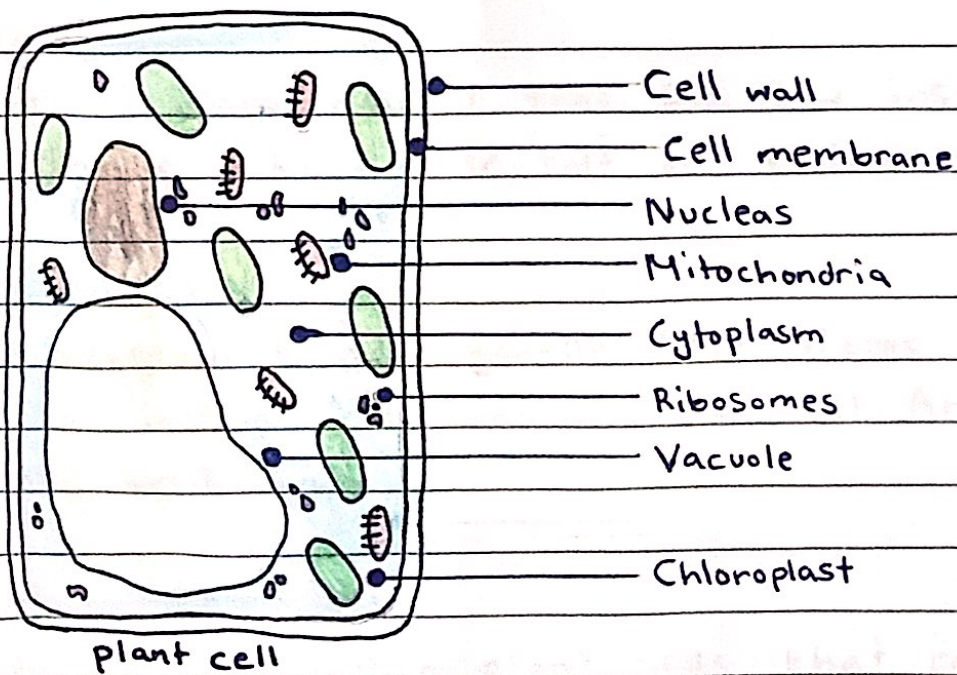
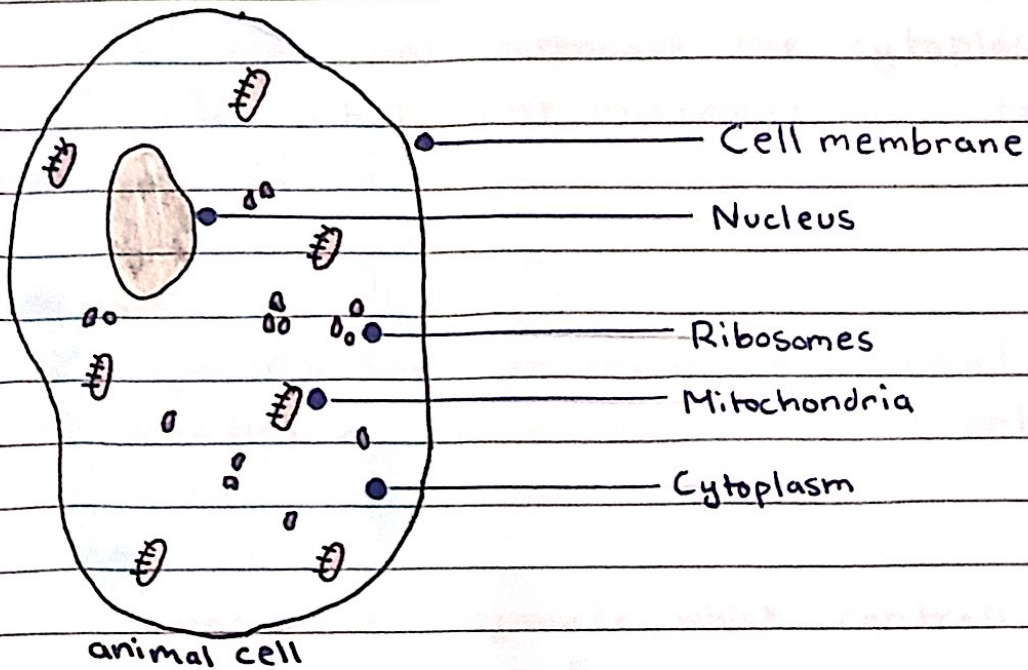




Cell membrane::

- the thin layer that surrounds the cytoplasm of the cell and controls the movement of materials into and out of the cell.

Cell wall::

- the rigid nonliving permeable wall that surrounds the membrane and encloses a plant cell.

Nucleus::

- a membrane-bound organelle which controls and regulates the activities of the cell.

Cytoplasm::

- the gelatinous liquid that fills the inside of a cell. It is composed of water, salt and various organic molecules.

Vacuole::

- a membrane-bound organelle that stores water, food and other nutrients required by the cell. Animal cells have rather small vacuoles.

Chloroplast::

- an organelle within plant cells that carries out photosynthesis with the help of chlorophyll.

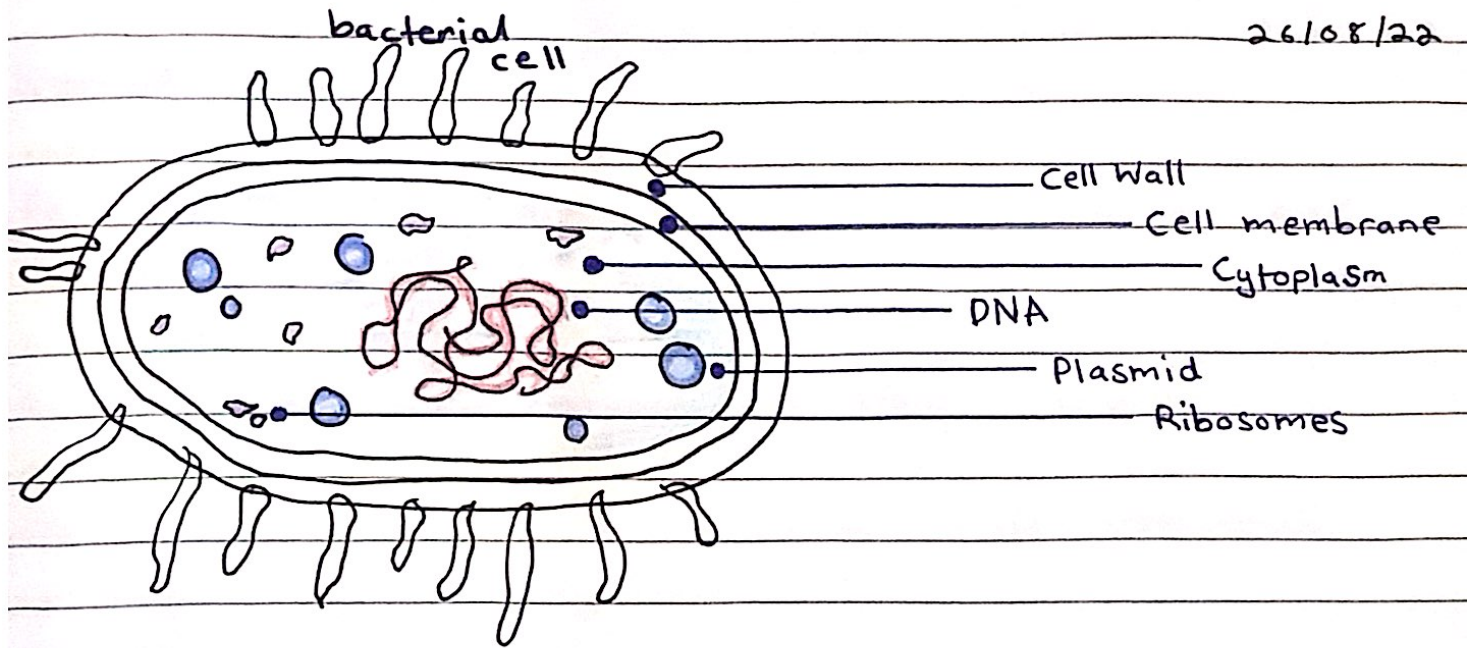
Mitochondria::

- the powerhouse of the cell that generates most of the chemical energy.

Ribosomes::

- the site of protein synthesis made of protein and RNA.

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DNA::

- The molecule inside cells that contains the genetic information responsible for the development and function of an organism.

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Specialised cells:

ciliated cells:

Facilitates the movement of mucus in the trachea and bronchi.

root hair cells:

surface area improves water absorption in plants.

palisade mesophyll cell:

closely packed together to maximise light absorption resulting in more photosynthesis

neurones:

they conduct electrical impulses

red blood cells:

absence of nucleus to store more oxygen for transportation.

sperm and egg cells (gametes):

Function of reproduction in the body.

Cell:

- the basic building blocks of all living things

Tissue:

- a group of cells with similar structure and function.

Organ:

- a collection of tissues that perform a particular function.

Organ system:

- a group of organs that work to accomplish a common goal.

Organism:

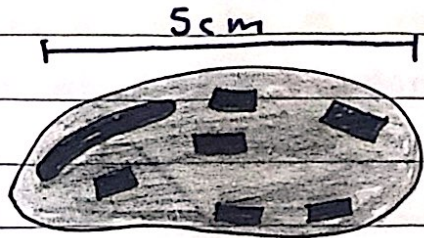
- a living thing made up of one or more cells

SIZE OF SPECIMENS

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Formula

magnification = image size $\div$ actual size



This is an image of a chloroplast. The image is 5 cm Long

The actual length is 5 μm (micrometer)

What is the magnification of the image?

cm $\rightarrow$ μm

$\times 10,000$

5cm $\rightarrow$ 50 000 μm

magnification = 50 000 $\mu\text{m} \div 5 \mu\text{m}$

magnification = $\times 10\,000$

mm $\rightarrow$ μm

$\times 1000$

cm $\rightarrow$ μm

$\times 10\,000$

$\mu\text{m} \rightarrow$ mm

$\div 1000$

$\mu\text{m} \rightarrow$ cm

$\div 10\,000$